



Simulation of the compaction of an all-solid-state battery cathode with coated particles using the discrete element method

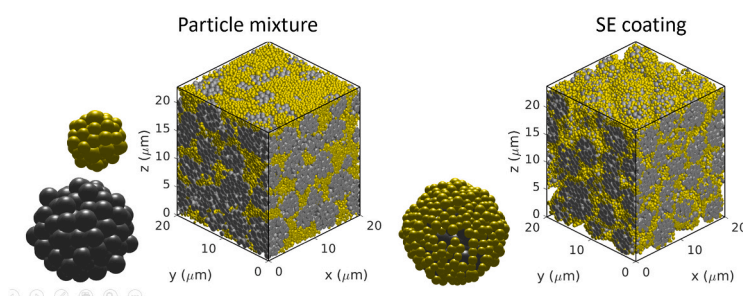
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HIGHLIGHTS

- Cold-pressing of all-solid-state battery (ASSB) electrodes was simulated by DEM.
- SE was simulated with two structures: free particles or as coated particles on AM.
- The damage of AM particles and the tortuosity of AM and SE was calculated.
- SE coating promotes effective ionic transport via the superior percolating network.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

All solid-state battery (ASSB)
Discrete element method (DEM)
Solid electrolyte delamination
Dual shell structure
Solid electrolyte coating
Percolation pathway

ABSTRACT

The simulation of a cold-pressed electrode, which consists of active material (AM) particles coated with a solid electrolyte (SE), of an all-solid-state battery (ASSB), is presented. The aim of this study is to elucidate the effect of the SE coating structure on the performance of the ASSB. The new method we developed to simulate the SE-coated AM particles is based on a discrete element method. This approach enables us to simulate mold compaction as well as the delamination of SE from the AM particles. Simulation of the AM as clustered particles held together by fusion bonding allows the mechanical damage to the AM to be reproduced simultaneously. For a ductile SE material, new fusion bonding contacts with other particles could develop when pressure is applied. Our simulations suggest that the SE coating promotes more effective ionic transport, in contrast to conventional particle mixture, owing to the superior percolating network. At low mold pressures, electron transport between the AM particles is disrupted because of the shielding effect of the SE coating. This effect vanishes at higher mold pressures. Both types of electrodes (particle mixture and SE coating) have similar damage profiles, with damage concentrated near the current collector.

1. Introduction

The market for electric vehicles (EVs) has grown enormously over the past decade, and the development of lithium-ion batteries (LIBs) is the main driving force behind their success. Despite significant

improvements in the service lifetimes and energy densities of LIBs, concerns regarding the short driving range, long charging time, and fire hazard associated with these batteries continue to prevail in the consumer market [1]. In all solid-state batteries (ASSBs), the organic liquid electrolyte is replaced by an inorganic solid electrolyte (SE). The core of

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<https://doi.org/10.1016/j.jpowsour.2022.231279>

Received 14 December 2021; Received in revised form 11 February 2022; Accepted 6 March 2022

Available online 18 March 2022

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an ASSB consists of an anode, a cathode, and an SE layer, where the latter acts as a separator between the anode and cathode. Both of these electrodes contain a particle mixture of SE and active material (AM). In comparison to LIBs, the driving range and charging time of EVs powered by ASSBs has the potential to be significantly improved, and the inorganic SE is inflammable, which drastically curtails the fire hazard.

ASSB technology has gradually improved over the last few decades, and in 2020, Toyota announced an ASSB with a capacity of 500 km that allows fast charging within 10 min [2]. In 2021, Hitachi Zosen Corp. developed a high-capacity ASSB suitable for use in industrial machinery and satellites [3]. However, for ASSBs to reach the broader consumer market their performance would have to be improved and their manufacturing costs lowered. A major reason for the lack of performance is the resistance of the solid-solid interface caused by loose contact, grain boundaries, and chemical and electrochemical reactions [4].

Ideally, a solid electrolyte for ASSBs is required to have high ionic conductivity, low electronic conductivity, wide electrochemical stability, and high processability. Potential candidates with these properties are oxide-, polymer-, and sulfide-type materials [5]. Oxide electrolytes have high ionic conductivity and excellent electrochemical stability. However, these electrolytes require high-temperature sintering, which may cause unwanted side reactions at the interface between the AM and SE [6].

Sulfide-type SEs are highly processable and deformable owing to their low Young's modulus of elasticity and low hardness, which makes it possible to fabricate electrodes at low temperatures and avoid the side reactions resulting from high-temperature sintering [7,8]. These SEs also have a high Li⁺ transference number, which is a deciding factor for restricting dendrite growth [9]. In particular, two groups of sulfide SEs have recently attracted attention: argyrodite and lithium-phosphate sulfides (LPSSs). Argyrodite-type sulfides have been used by Samsung to develop high-capacity and long-lasting batteries [10]. LPSSs are highly ductile (Pugh's ratio ~2.4 [11]), and can be produced in the form of fine powders by ball milling and amorphization [12]. LPSSs can be combined with active material (AM) particles to produce a compact electrode by cold pressing at room temperature [8,13].

Despite their advantages, sulfide electrolytes have poor mechanical and chemical interfacial stabilities, which increase the risk of damage and adversely affect charge transport [14]. In addition, the desire is to increase the charge capacity of ASSBs by increasing the amount of AM. However, this requires the SE to be highly connected, even at low SE fractions. The integrity and the charge transport performance was demonstrated to improve by producing an electrode with a more uniform distribution of AM and SE particles [15,16]. Alternatively, the AM particles were coated with SE before pressing to create a sufficient ion-conducting path even at low SE fractions [17]. The SE coating was added in addition to the antioxidizing layer, usually consisting of LiNbO₃, and this combination is often referred to as having a dual-shell structure. In this regard, various approaches have been adopted, including pulsed laser deposition of LPS@LiCoO₂ [18], a solution method for LiI-Li 4-SnS₄@LiCoO₂ [19], atomic layer deposition of lithium niobium oxides (LNO) [20], wet dispersion of Li₁₀GeP₂S₁₂@LiCoO₂ [21], and dry coating of LPS@NCM [17].

Nakamura et al. [17] demonstrated that a half-cell with a dual-shell structure had improved rate and cycling performance in comparison with a half-cell fabricated using the ball-milling method. Reconstruction from FIB-SEM images and tortuosity calculations, confirmed that the dual-shell structure offers a larger surface area for AM-SE contact and an enhanced SE percolation network. The dry coating method also has the advantage of being much less demanding in terms of energy consumption than pulsed laser deposition. The results of our simulation study are compared with those of the experimental study by Nakamura et al. [17] as a reference. In contrast to the reference study, which uses a top-down approach via FIB-SEM image reconstruction, we use a bottom-up approach with particle dynamics. This approach enables us to

crosscheck the results to gain a deeper understanding of the physical interactions.

The simulation of ASSBs is expected to lead to a more in-depth understanding of the interactions and the cause-effect relationship between their components, and could serve to guide ASSB electrode design to improve the performance. Various top-down models of ASSBs have been reported in the literature. Fathiannasab et al. [22] used a transmission X-ray microtomography system to elucidate the ASSB structure, particulars of which were input into a simulator to calculate the galvanostatic discharge. The results showed that electrodes fabricated at lower mold pressures have more voids that hinder ionic transport. Yu et al. [23] developed a smoothed boundary method to solve the deformation and stresses in an ASSB electrode and found that the thickness of the electrode has a significant impact on the bending deformation.

Several bottom-up particle-based models have been reported in the literature. Bielefeld et al. evaluated the electron transport of electrode particles and related the findings with the percolation theory. The electron and ionic conductivities may drop to zero if the particles are unable to form a connecting chain [24,25]. The discrete element method (DEM) is a simulation method that solves Newton's laws of motion of moving particles and is usually employed to study rock mechanics and bio-reactors [26]. A packing DEM model was developed by Sangros Gimenez et al. to estimate the electronic conductivity of AM and carbon black (CB) particles [27]. Only the AM and CB particles were resolved in the simulation, and the space occupied by the SE was assumed to be void space between these particles. A packing model based on the DEM was developed by Shi et al. [28] to model the charge transport, which was concluded to improve provided the ratio of the AM diameter to the SE diameter is large.

We previously developed a 2D model for ASSB fabrication and cell intercalation. The model was able to simulate plastic deformation by modeling the equilibrium overlap distance [29]. The model was subsequently extended to three dimensions to estimate the effect of the fabrication mold pressure on ionic transport [30]. Our results for cold pressing agreed well with the experimental results obtained for the ball-milled SE. We derived a new relationship for the electrochemical conductivity of a mold pressed electrode with sulfide electrolyte. However, it was discussed that the previous model lacked the ability to simulate AM breakage during mold pressing.

The aim of the study reported here is to gain insight into the effect of depositing the SE as a coating on the cathode AM particles on charge transport and mechanical damage resulting from mold pressure when fabricating the electrode. To this end, we used DEM simulations and simulated the SE coating as DEM particles attached to the AM. By modeling the AM particles as aggregates of primary particles, we can simulate the breakage of fusion bonding contacts between particles and thus the aforementioned damage to the AM.

In this study, we aim to further understand the extent to which the SE shell structure affects the ionic and electron transport in relation to networks of percolating particles. In terms of ionic transport, we are particularly interested in the relationship between the improved ionic performance of the SE, as observed by Nakamura et al. [17], and the coated structure, as well as the relationship between the observed lower AM conductivity and the shielding of pathways by the SE coating. Furthermore, we aim to determine whether the SE coating provides a damping effect that might limit the damage to the AM particles. The simulation method of coated SE on AM particles is a novel technique that has not been previously demonstrated with DEM. In addition, the results describing the damage distribution of AM particles are novel. We have further improved the accuracy of our compaction model, and detailed descriptions of the methods can be found in the corresponding MethodsX article [31].

2. Methods

2.1. Overall description of our approach

The DEM simulations were used to construct a number of different ASSB electrodes using different mold pressures and two different approaches to fabrication: particle mixture and a SE coating. All simulations were conducted using our developed MATLAB code, which is a slight modification of the code in a previously published paper [30]. The simulations were conducted on a workstation with an Intel® Xeon® W-2145 CPU (3.70 GHz, 8 cores). The duration of the computation for each data point included in the simulation was approximately 4 h.

2.2. Simulation of the particle aggregates

A SEM image of an AM aggregate consisting of primary NCM particles is shown in Fig. 1a. To obtain similar aggregates for our DEM model (Fig. 1b) we used an approach developed in an earlier study. The simulation of the aggregate reconstruction mimics the interactions that occur during annealing, where fusion bonding develops between the primary particles to form aggregates [32,33]. To control the overlap between particles, a shoving algorithm was applied [34]. The overlap distance was controlled to have a maximum value of $0.6 R_{\text{eff}}$ where R_{eff} is the effective radius (see equation (3)). As shown in Fig. 1a, the particle mixture simulation included both AM and SE particles as spherical aggregates. The aggregate simulation was conducted individually so that the shape and size varied for each aggregate. The primary particle diameters ($d_{\text{AM}}^{(\text{pp})}$ and $d_{\text{SE}}^{(\text{pp})}$) and aggregate diameters ($d_{\text{AM}}^{(\text{agg})}$ and $d_{\text{SE}}^{(\text{agg})}$) are indicated in Table 1. These were obtained from the experimental results of Nakamura et al. [17], and we observed through analysis that the number distribution of AM aggregates was closely lognormal with a modal peak near $5 \mu\text{m}$ and had a lognormal standard deviation of 0.2. The number distribution of the SE diameter for the particle mixture had a peak near $0.5 \mu\text{m}$, but it had a significant tail that became dominant with respect to the volume (up to 70%). Therefore, we assigned $d_{\text{SE}}^{(\text{pp})}$ to $0.5 \mu\text{m}$ and $d_{\text{SE}}^{(\text{agg})}$ $1.5 \mu\text{m}$ by size distribution fitting. The lognormal standard deviation of the primary SE particles and SE aggregates was set to 0.3 according to our fitting. To simulate the particle mixture, we set the volume fraction of the non-aggregated SE to 30%.

Table 1

Properties for AM (NCM) and SE (LPS) in the simulations.

Symbol	Value	Units	Description	Reference
$d_{\text{AM}}^{(\text{agg})}$	5	μm	AM aggregate mean diameter	Nakamura et al. [17]
$d_{\text{AM}}^{(\text{pp})}$	1	μm	AM primary particle mean diameter	See Section 2.2
$d_{\text{SE}}^{(\text{agg})}$	1.5	μm	SE aggregate mean diameter	See Section 2.2
$d_{\text{SE}}^{(\text{pp})}$	0.5	μm	SE primary particle mean diameter	See Section 2.2
e	0.5	–	Restitution coefficient	This study
E_{AM}	199	GPa	AM Young modulus (NCM)	Cheng et al. [37]
E_{SE}	24	GPa	SE Young modulus (LPS)	Sakuda et al. [8]
f_{AM}	0.734	–	AM volume fraction within solid	Nakamura et al. [17]
H_{AM}	11.2	GPa	Hardness of AM (NCM)	Cheng et al. [37]
H_{SE}	1.9	GPa	Hardness of SE (LPS)	McGrogan et al. [7]
μ	0.5	–	Friction coefficient	This study
ν_{SE}	0.32	–	SE Poisson ratio (LPS)	Sakuda et al. [13]
ν_{AM}	0.25	–	AM Poisson ratio (NCM)	Cheng et al. [37]

2.3. Simulation of coating method

The SE coating on the AM was simulated before the AM particles were placed in the computational domain. A SEM image of a SE coated AM particle is shown in Fig. 1c and can be compared with that of our model in Fig. 1d. The SE coating simulation first constructed the AM aggregates as described in section 2.2, but differed in that the coating was applied by attaching SE particles onto the AM surface. First, the location of the surface was numerically estimated. This was accomplished by first performing a 3D Delaunay triangulation of the AM subparticle positions of each main AM aggregate. This resulted in triangulation consisting of tetrahedrons with corners at the subparticle centers. The surface was roughened by removing tetrahedrons with large edge lengths near the surface from the triangulation, step by step until no long edges remained on the triangulation surface. The SE particles were randomly placed on this surface; then the position of these coated particles was recalculated using the shoving algorithm described in Subsection 2.2.

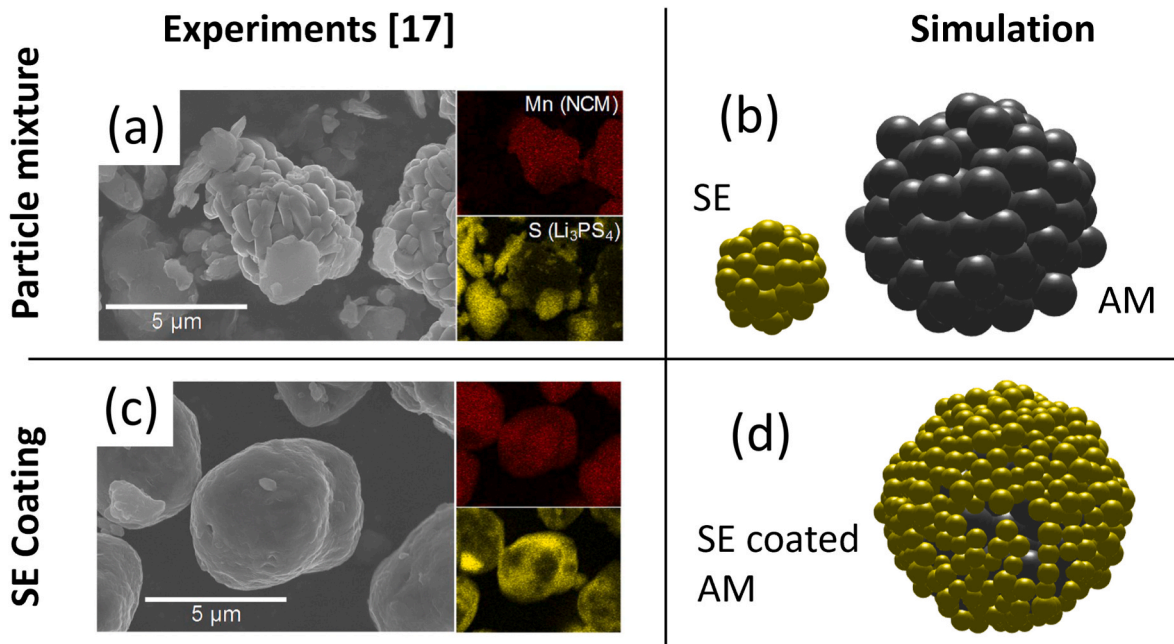


Fig. 1. Experimental and simulation results using the particle and coating methods. The SEM images to the left were reproduced from Nakamura et al. [17].

2.4. Simulation of cold pressing with DEM

The cold pressing simulation was initiated by placing the coated and/or uncoated aggregates in a computational domain that represented the mold. This placement was performed in two steps. First, the positions of the aggregates were randomly set. Second, by slightly moving the aggregates by the shoving algorithm explained in Section 2.2, the overlap between the aggregates was eliminated. During the cold pressing simulations, the top wall moved downwards and was controlled by the pressure of the contacting particles. Owing to the pressure, stress builds up between particles, which first undergo elastic deformation. As the pressure increases further, the stresses eventually increase beyond the particle hardness, and plastic deformation occurs.

The plastic deformation was quantified by the equilibrium overlap distance between two particles. We separated the plastic deformation into two modes: sintered and non-sintered where the former results in development of new fusion bonds between the particles. As elaborated by Sakuda et al. [8], LPS allow for room temperature sintering where fusion bonds can develop by pressure alone. However, this sintering process is not feasible for NCM at room temperature but AM–AM fusion bonds exist initially within AM aggregates due to high temperature sintering prior to the cold pressing. Fusion bonds between the particles can be separated if the tensile stress exceeds the hardness of the material. Fusion bond separation is equivalent to cracking damage within the AM and SE aggregates, and can also describe the delamination of the SE from the AM surface. A detailed description of our developed deformation model can be found in our MethodsX article [31].

The wall velocity was controlled using numerical PID control. The PID error function was based on the difference between the current and set pressure, and eventually, the velocity reached a point at which the pressure remained constant. Periodic boundary conditions were applied using the x- and y-coordinates, and the bottom wall was fixed.

2.5. Equations used in the DEM simulations

We calculated the particle dynamics from Newton's laws of motion and considered both elastic and plastic interactions. In the case of the absence of plastic deformation, the elastic spring force between the two particles is given by

$$F_{\text{spring}} = k_n h_{\text{ov}} \quad (1)$$

where h_{ov} is the overlap distance and k_n is the spring coefficient. We used the Hertzian contact law, for which k_n is given by

$$k_n = 4 / (3 \cdot E_{\text{eff}} h_{\text{ov}}^{1/2} R_{\text{eff}}^{1/2}) \quad (2)$$

where R_{eff} [m] is the effective radius of particles i and j

$$R_{\text{eff}}^{-1} = R_i^{-1} + R_j^{-1}. \quad (3)$$

The effective elastic modulus E_{eff} is given by

$$E_{\text{eff}}^{-1} = (1 - \nu_i^2) E_i^{-1} + (1 - \nu_j^2) E_j^{-1} \quad (4)$$

where ν is Poisson's ratio. To include plastic deformation, we introduced the equilibrium overlap h_{eq} into Equation (1) as follows:

$$F_{\text{spring}} = k_n (h_{\text{ov}} - h_{\text{eq}}). \quad (5)$$

The new parameter h_{eq} represents the plastic deformation displacement between two particles. The equilibrium overlap was simulated using the following rate expression:

$$\frac{\partial h_{\text{eq}}}{\partial t} = \begin{cases} \frac{F_{\text{spring}} - F_{\text{th}}}{t_{\text{rel}} k_n}, & \text{if } F_{\text{spring}} > F_{\text{th}} \\ 0, & \text{if } -F_{\text{th}} < F_{\text{spring}} < F_{\text{th}} \\ \frac{F_{\text{spring}} + F_{\text{th}}}{t_{\text{rel}} k_n}, & \text{if } F_{\text{spring}} < -F_{\text{th}} \end{cases} \quad (6)$$

where t_{rel} is the relaxation time. Positives values of $\partial h_{\text{eq}}/\partial t$ indicates consolidation and negative particle detachment. The threshold force F_{th} is modeled as follows:

$$F_{\text{th}} = \min(H_i, H_j) A_{\text{con}} \quad (7)$$

where H is the hardness of the particle material and A_{con} is the contact area. During detachment the value of F_{th} is set to zero if the particle contact is missing fusion bonds. This applies to AM–AM contacts that are not within the same aggregate or broken pre-sintered contacts. We also introduced correction factors for A_{con} and k_n at large deformations to increase the accuracy when the porosity approaches zero. More details can be found in our MethodsX article [31].

Table 1 lists the key parameters used in the simulations, most of which were obtained from the literature. In addition to these equations, the static/sliding friction force (with friction coefficient μ), collision damping (with restitution coefficient e), and wall forces were considered during the cold pressing simulation. We used the velocity Verlet scheme [35] for the numerical time integration. For the tortuosity calculations, we mapped the particles onto a grid [30] and used the MATLAB app TauFactor [36] for the calculations.

3. Results

3.1. Particle distribution

The simulated and experimental spatial distributions of the AM and SE are shown in Fig. 2 at a mold pressure of 360 MPa. The 3D particle plot (Fig. 2a) shows the simulation results obtained with the particle mixture method where the SE particles are colored by individual aggregates. The pressing direction is along the z-axis, which is perpendicular to the current collector boundary at the bottom and the SE boundary at the top. A slice of the same simulation results is shown in Fig. 2b, and can be compared with the experimental results of Nakamura et al. [17] in Fig. 2c. The simulated morphology was in good agreement with the experimental results in terms of appearance. Similarly, the results of the simulations using the coating method are shown in Fig. 2 (d–f). Note that, because the SE was applied in the form of a coating, the SE material is concentrated as a film around the AM particles.

3.2. Volume fractions and contact area

The volume fractions of the different components (AM, SE, and pores) are plotted in Fig. 3a, and are in good agreement with the FIB-SEM experimental results. The specific surface areas of the interfaces are shown in Fig. 3b. Good agreement was obtained for the AM-SE surface areas for the SE coating simulation, whereas the simulated areas of the other interfaces were significantly larger. We elaborate on the reasons for this in Subsection 4.3.

3.3. Aggregates and percolation clusters

The number of AM aggregates is plotted as a function of the mold pressure in Fig. 4a. For the particle mixture simulations, the number of aggregates starts to increase at 50 MPa. For the SE coating simulations, at low mold pressures up to 200 MPa, the number of aggregates initially remains essentially the same, with a small increase that can be observed. The increase in the number of AM aggregates is attributable to a small number of fragments resulting from structural damage to the AM, which

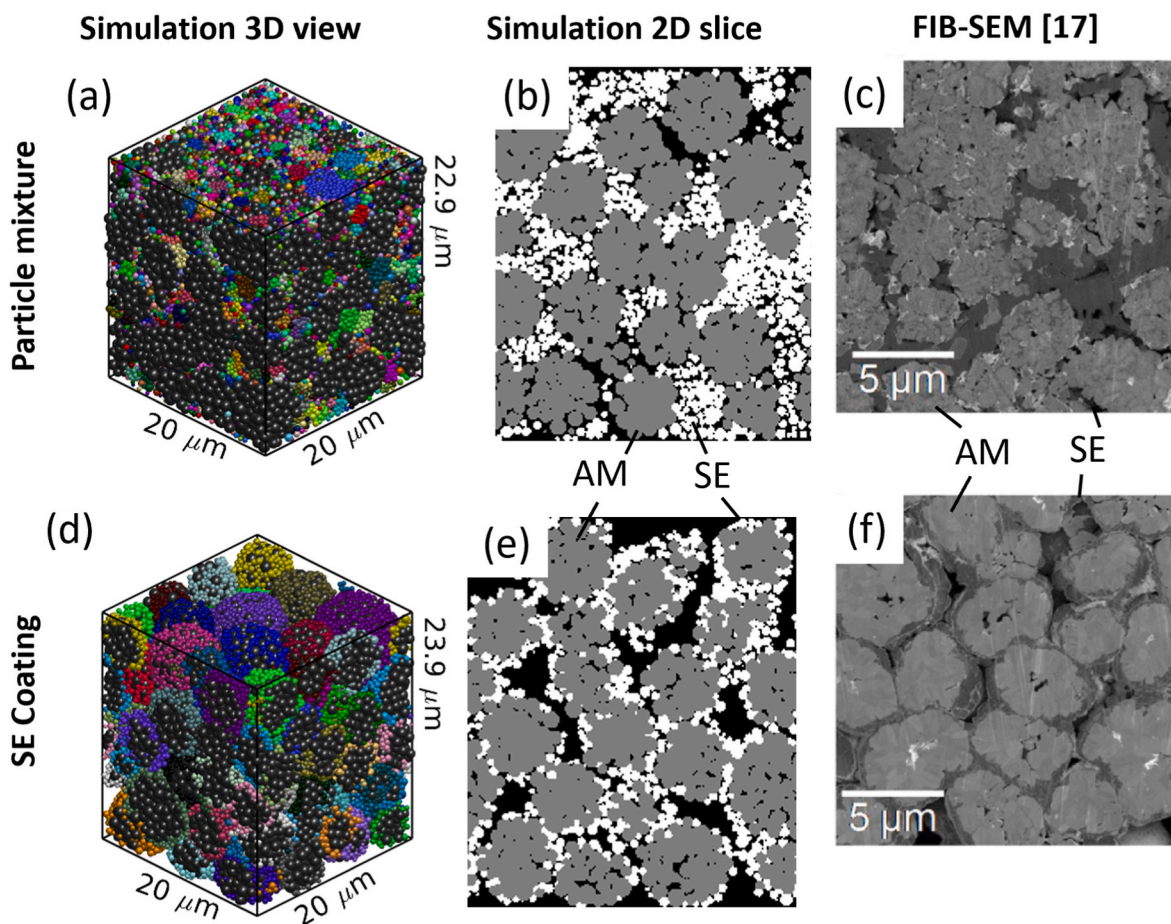


Fig. 2. Comparison of the simulation and experimental results. Particle distribution after cold pressing at 360 MPa using both the particle mixture (a, b, c) and SE coating (d, e, f) methods. The experimental results in (c) and (f) were reproduced from Nakamura et al. [17].

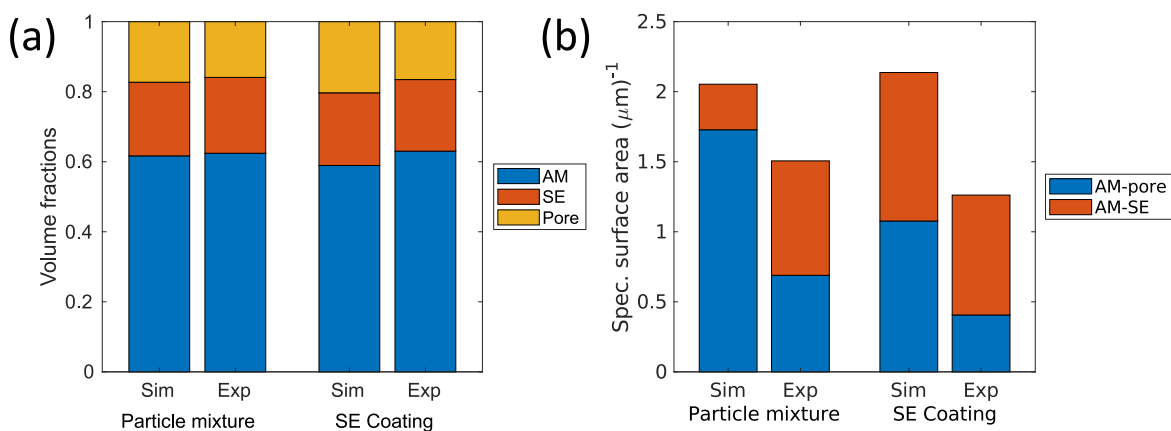


Fig. 3. Volume fractions and interface surface areas.

we explore in more detail in Section 3.4. The aggregates can be connected with each other by contact forces to form larger percolation clusters, which are plotted in Fig. 4b. Initially, at low mold pressures, the number of percolation clusters is equal to the number of aggregates, but this number quickly decreases with mold pressure as the aggregates become connected. The reduction in the number of AM clusters is much faster with the particle mixture method than with the SE coating method. This is related to the way in which contacts between AM-AM particles develop as a result of delamination of the SE layer, which is considered in the next section.

The relative AM percolation volume is the relative volume of the AM that participates in electron transport. The relative percolation volume of AM is plotted in Fig. 4c, which shows that the AM for the particle mixture is almost fully percolated at 25 MPa, whereas 100 MPa is required for the coating method. On the other hand, in the case of the SE percolation volume, the particle and coating methods produce the opposite results, as shown in Fig. 4d. Specifically, the relative SE percolation volume with the coating method exceeded 90% at 25 MPa, whereas mold pressure of 360 MPa was required with the particle mixture to attain the same percolation volume.

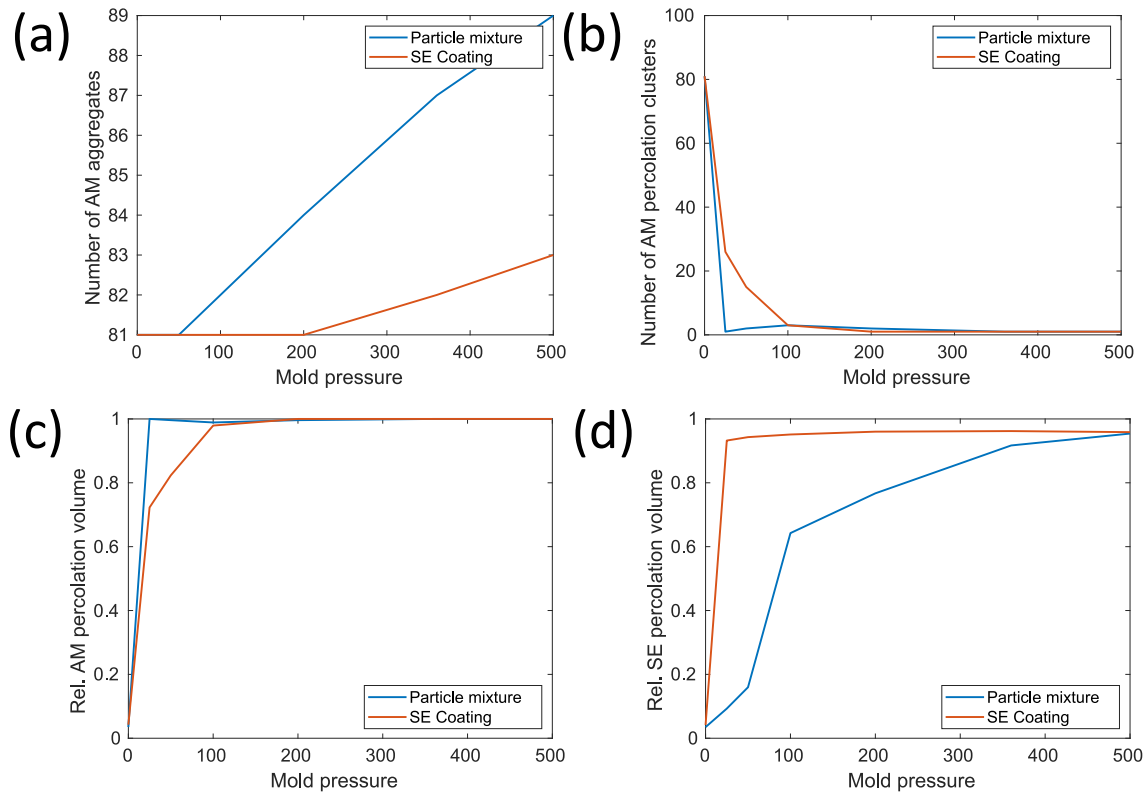


Fig. 4. Comparison of the two different simulated fabrication methods with respect to percolation. Number of AM aggregates (a), number of AM percolation clusters (b), and relative percolation volume of AM (c) and SE (d).

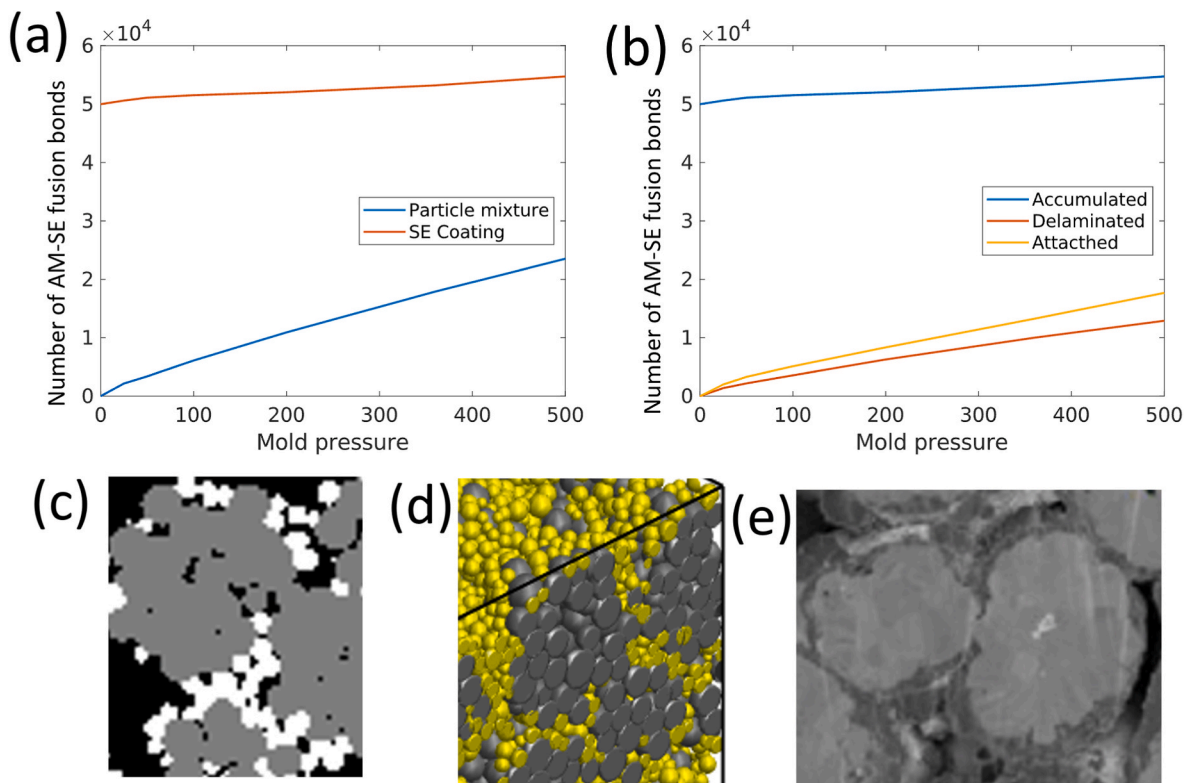


Fig. 5. Number of AM-SE fusion bonds (a) and the balance for the coating method (b). Images of the delaminated regions: 3D simulation (c), simulation slice (d) and experimental results (e). Subfigure (e) was reproduced from Nakamura et al. [17].

3.4. Delamination of SE coating and reattachment

The number of AM-SE fusion bonds is plotted in Fig. 5a versus the mold pressure. The results obtained with the simulated SE coating and particle mixture differ markedly. This was expected, as AM-SE fusion bonds are present in the coating method from the start. The largest difference occurs at low mold pressures, but the difference at high mold pressures is also highly significant.

The degree of delamination and attachment of the coating layer was investigated by conducting further analysis, as shown in Fig. 5b. The parameter $n_{\text{accumulated}}$ is the current number of AM-SE fusion bonds and is related to the other values as follows:

$$n_{\text{accumulated}} = n_{\text{initial}} + n_{\text{attached}} - n_{\text{delaminated}} \quad (8)$$

where n_{initial} is the number of initial contacts before the pressing step in the simulation, $n_{\text{delaminated}}$ is the number of delaminated contacts, and n_{attached} is the attached number of fusion bonds. The number of delaminated AM-SE particles is significant; however, the number of attached AM-SE fusion bonding contacts increases to a greater extent, resulting in a net accumulation as the mold pressure increases. High-magnification views of the simulated results in Fig. 5c and d and the experimental results in Fig. 5e of the coating method reveal regions of delaminated SE at the contact point between pairs of AM particles.

3.5. Mechanical damage

In contrast to contacts that include SE, we did not allow new AM-AM fusion bonds to develop during cold pressing, because the AM consisting of NCM requires high-temperature sintering for such bonds to develop.

The number of AM-AM fusion bonds is plotted in Fig. 6a as a function of the mold pressure. These results show that the number of AM-AM fusion bonding contacts decreases significantly with the mold pressure and these data enable the AM damage to be quantified. We defined our parameter to express the AM damage as the relative decrease in the AM-AM fusion bonding contact in relation to the initial value as follows:

$$f^{(\text{AM})} = \frac{n_{\text{initial}}^{(\text{AM-AM})} - n_{\text{current}}^{(\text{AM-AM})}}{n_{\text{initial}}^{(\text{AM-AM})}} \quad (9)$$

The overall damage was plotted versus the mold pressure in Fig. 6b. As expected, the AM damage increased with the mold pressure up to a value of approximately 12% at 500 MPa. The overall damage for the particle and coating method is similar, and the small differences between them are not likely to be statistically significant. In Fig. 6c, the AM damage parameter was averaged over each AM particle. The most damaged particles were found close to the upper and lower wall boundaries. In Fig. 6d, the AM damage parameter is averaged over a horizontal cross section and is plotted against the vertical position. These results show that, for both the particle and coating methods, the damage was concentrated near the upper and lower boundaries.

3.6. Charge transport

The charge transport of electrons and Li ions depends on the tortuosity factors of the AM and SE and these results are shown in Fig. 7. The tortuosity factors for the SE are plotted against the mold pressure in Fig. 7a. The dashed lines are simulations with periodic boundary conditions (PBC). The simulation results without PBC (solid lines) are in good agreement with the FIB-SEM experimental results (triangles) from

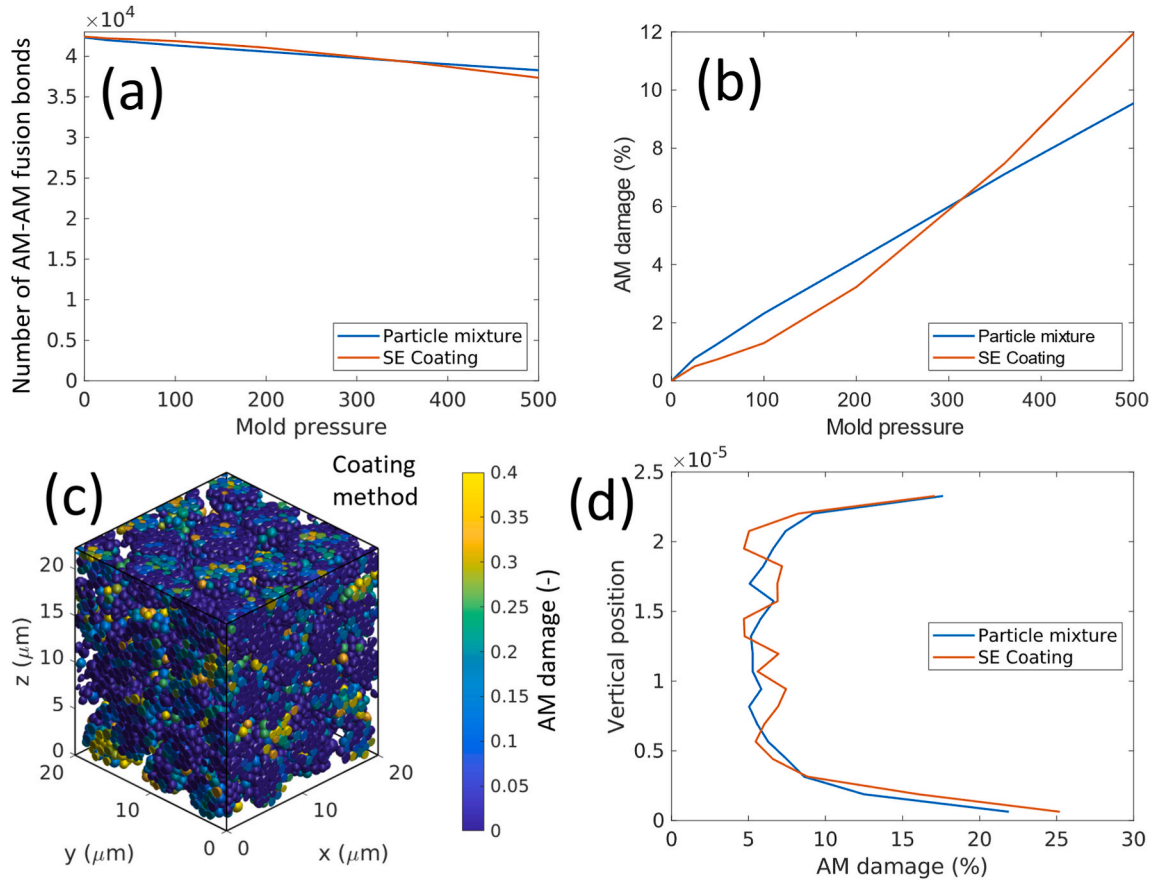


Fig. 6. Damage inflicted upon the AM in the electrode as a result of mold pressure. Number of AM-AM fusion bonds (a) and the AM damage (b) plotted versus the mold pressure. Distribution of damage across the particles (c) and along the vertical axis (d).

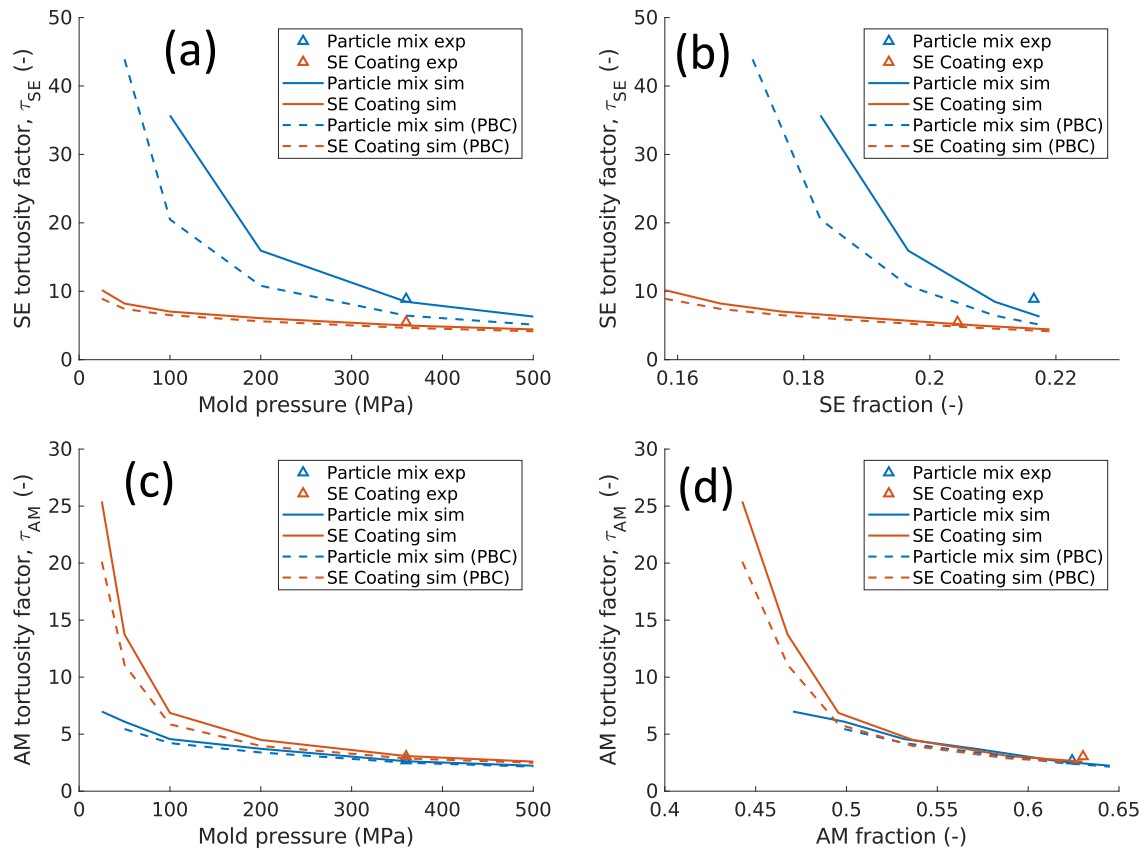


Fig. 7. Tortuosity factors for the SE (a, b) and AM (c, d) from the simulation and experiments (Nakamura et al. [17]).

Nakamura et al. [17].

The coating method (red lines) resulted in lower tortuosity than the particle mixture (blue lines). This also holds true even when the data are plotted with the SE volume fraction (Fig. 7b). The differences between the fabrication methods are reversed for AM tortuosity, which is plotted as a function of the mold pressure in Fig. 7c. At lower mold pressures, the coating method resulted in significantly larger tortuosity. However, at higher mold pressures, the difference almost completely disappeared. The AM tortuosity factor exhibited the same tendency in response to changes in the fraction of AM (Fig. 7 d).

4. Discussion

4.1. Novelty and importance of the study

This study led to the development of a novel method to simulate the compaction of an ASSB electrode with particles that have a dual-shell structure. Our method is based on two earlier studies of ASSBs, where compaction as a result of plastic deformation was simulated by modeling the equilibrium overlap between particles. We simulated the ball-milling process by simulating the SE as aggregates, as in our earlier studies. The particles within the aggregate were considered to separate if the tensile stress exceeded the hardness of the material. In addition, based on the ductility of the SE, new fusion bonds were allowed to develop upon high compression.

This study included two new methods: (i) simulation of AM particles covered by SE coating and (ii) simulation of the AM as aggregates. In our simulation of the SE coating, we computed the degree of SE delamination and reattachment during cold pressing. Simulation of the AM particles as aggregates with breakable fusion bonds enabled us to estimate the degree of damage based on the mold pressure and location of the particles in the electrode.

4.2. Summary of the results

The SE coating method is able to substantially improve the effective conductivity of the SE in comparison to particle mixture. This is because of the less tortuous pathways and the superior connections of the percolating network for ionic transport. The tortuosity of the AM network was higher for the SE coating method, which is in agreement with the experimental results of Nakamura et al. [17] At low mold pressures, the SE coating caused the percolation paths of the AM to separate; however, at mold pressures exceeding 200 MPa, the AM particles formed a fully percolated network.

One of the reasons that a fully percolated network of AM particles could be obtained with the SE coating method is that delamination of the coated SE occurred at the point at which the AM particles made contact. The delamination opened new paths that allowed electrons to travel between core-shell structures. As delamination proceeded, new SE particles became attached to AM particles. Consequently, the number of newly attached particles was greater and their combined surface area was larger than those of the peeled particles, resulting in an overall increase in the contact surface area between AM and SE as the mold pressure increased. The particle density of the SE forming the shell of each core-shell structure increased, thereby leaving space for the AM particles to remain in contact.

However, the SE coating and particle mixture methods differed significantly in terms of percolation and charge transport behavior, whereas the difference in damage distribution was insignificant. According to our simulations, the coating layer offered the AM minimal protection against damage. The damage distribution profile indicated that the damage was mostly confined to areas near the wall boundaries, which resemble the boundaries of the current collector and the SE layer.

4.3. Shortcomings and future work

In this study, we evaluated the effect of the coating method on the tortuosity of AM and SE only within the cathode. However, for the overall performance, charge transport through the SE and cathode layers also plays a role. Moreover, we did not simulate the effect of the solid–solid interface on the performance, which plays a crucial role in the performance of ASSBs. In particular, the quality of the solid–solid interface can significantly alter performance [4]. Furthermore, we did not simulate coupled reactions and transport. Such a simulation would have been useful for evaluating the effect of SE and AM tortuosity on the performance.

As evaluated by Shi et al. [28], the size ratio of the AM to SE plays a crucial role in compaction and the ability to form a percolation path. Complemented simulations with different AM and SE sizes can be conducted in the future to evaluate the effect of the particle size on the performance.

The domain size (20 μm) is not large, considering the AM aggregate size (5 μm); therefore, we cannot rule out the aggregate number dependence on our results. However, considering the primary size of the particles (0.5–1 μm), it was difficult to increase the domain size further; it took 10 days to complete all the current simulations.

Although the simulations could accurately predict the volume fractions and surface area of the AM–SE interface, the AM–pore area was overestimated. The reason is that the simulation gives rise to many smaller internal pores within the AM in comparison to the experiments. In addition, the SE coating is less smoothly deposited in the simulations, which leaves the surfaces of the AM particles partly exposed.

Because our model focused on replicating the results of Nakamura et al. [17], who performed experiments with a half-cell, we conducted our simulations without an SE layer on top of the electrode. This means that the ductility of the LPS would have significantly changed the damage profile near the SE layer, and the damage at the top boundary is likely to have been lessened significantly. However, the extensive local damage at the current collector is realistically reproduced, as this component often consists of hard materials. To adjust the model such that it represents a real battery cell as closely as possible, we plan to include a top layer of SE particles in our future simulations.

We were able to simulate the delamination and reattachment of SE particles from AM. However, investigating the delamination of the antioxidating layer would also be of considerable interest. In the future, we plan to introduce additional particles that represent the antioxidating layer material.

Our model focused on the damage caused by AM during the fabrication process. We are currently testing our model to simulate expansion as a result of intercalation, the simulation of which would enable us to gain an understanding of damage development during battery operation.

5. Conclusions

A new approach to simulate cold pressing of SE-coated AM particles was developed. The proposed approach is based on our previously developed ASSB model that includes both elastic and plastic deformation. In this study, two new features were introduced to the model: the simulation of an AM covered by an SE coating and that of AM particles as a breakable cluster. These two new methods allowed us to simulate the delamination of SE from the coated layer and the mechanical damage inflicted upon the AM particles because of mold pressure.

The results of the DEM simulations agree well with the experimental results of Nakamura et al. [17] in terms of the volume fractions and tortuosity. When considering charge transport alone, the electrode simulated with the coating method had superior ionic effective conductivity owing to its well-developed SE percolation network. At lower mold pressures, the coating method lowered the electron transport performance, partly owing to the shielding effect of the coated SE. On

the other hand, at higher mold pressures, the electron mass transport through the AM coated with SE and the non-coated AM did not differ significantly. We attributed this to the delamination of the SE coating layer at the contact points between the AM particles. The difference in the AM damage profile was not significant for the coating and particle fabrication methods. However, the vertical damage profile was not constant, and the more extensive damage was concentrated near the current collector.

To evaluate the effect of the coating structure on the performance, we restricted our simulation to rough spherical AM structures, which were first placed randomly and then compressed. We did not simulate a large variation in AM shapes or AM ordered in any structured configuration. Therefore, we cannot make conclusions that are valid beyond particle-dispersed AM or with AM particles with shapes that largely differ from the spherical or elliptical.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by JSPS KAKENHI Grant-in-Aid for Scientific Research on Innovative Areas, “Science on Interfacial Ion Dynamics for Solid State Ionics Devices, Computational & Data Science” (Gp-A03, grant number 19H05815); and by MEXT “Program for Promoting Researches on the Supercomputer Fugaku” (Fugaku battery & Fuel Cell Project), grant number JPMXP1020200301.

References

- [1] U.S. Department of Energy, Batteries, Annual Progress Report, 2018, 2019.
- [2] Toyota's game-changing solid-state battery en route for 2021 debut, Nikkei Asia. (n.d.). <https://asia.nikkei.com/Spotlight/Most-read-in-2020/Toyota-s-game-changing-solid-state-battery-en-route-for-2021-debut> (accessed April 19, 2021).
- [3] “World's highest-capacity” solid-state battery developed in Japan, Nikkei Asia. (n.d.). <https://asia.nikkei.com/Business/Energy/World-s-highest-capacity-solid-state-battery-developed-in-Japan> (accessed December 5, 2021).
- [4] A. Banerjee, X. Wang, C. Fang, E.A. Wu, Y.S. Meng, Interfaces and interphases in all-solid-state batteries with inorganic solid electrolytes, *Chem. Rev.* 120 (2020) 6878–6933, <https://doi.org/10.1021/acs.chemrev.0c00101>.
- [5] A. Manthiram, X. Yu, S. Wang, Lithium battery chemistries enabled by solid-state electrolytes, *Nat. Rev. Mater.* 2 (2017) 1–16, <https://doi.org/10.1038/natrevmats.2016.103>.
- [6] K. Park, B.-C. Yu, J.-W. Jung, Y. Li, W. Zhou, H. Gao, S. Son, J.B. Goodenough, Electrochemical nature of the cathode interface for a solid-state lithium-ion battery: interface between LiCoO₂ and garnet-Li₇La₃Zr₂O₁₂, *Chem. Mater.* 28 (2016) 8051–8059, <https://doi.org/10.1021/acs.chemmater.6b03870>.
- [7] F.P. McGrogan, T. Swamy, S.R. Bishop, E. Eggleston, L. Porz, X. Chen, Y.-M. Chiang, K.J.V. Vliet, Compliant yet brittle mechanical behavior of Li₂S–P₂S₅ lithium-ion-conducting solid electrolyte, *Adv. Energy Mater.* 7 (2017) 1602011, <https://doi.org/10.1002/aenm.201602011>.
- [8] A. Sakuda, A. Hayashi, M. Tatsumisago, Sulfide solid electrolyte with favorable mechanical property for all-solid-state lithium battery, *Sci. Rep.* 3 (2013) 1–5, <https://doi.org/10.1038/srep02261>.
- [9] X. Zhang, A. Wang, X. Liu, J. Luo, Dendrites in lithium metal anodes: suppression, regulation, and elimination, *Acc. Chem. Res.* 52 (2019) 3223–3232, <https://doi.org/10.1021/acs.accounts.9b00437>.
- [10] Y.-G. Lee, S. Fujiki, C. Jung, N. Suzuki, N. Yashiro, R. Omoda, D.-S. Ko, T. Shiratsuchi, T. Sugimoto, S. Ryu, J.H. Ku, T. Watanabe, Y. Park, Y. Aihara, D. Im, I.T. Han, High-energy long-cycling all-solid-state lithium metal batteries enabled by silver–carbon composite anodes, *Nat. Energy* 5 (2020) 299–308, <https://doi.org/10.1038/s41560-020-0575-z>.
- [11] A. Kato, M. Nose, M. Yamamoto, A. Sakuda, A. Hayashi, M. Tatsumisago, Mechanical properties of sulfide glasses in all-solid-state batteries, *J. Ceram. Soc. Japan* 126 (2018) 719–727, <https://doi.org/10.2109/jcersj2.18022>.
- [12] A. Hayashi, S. Hama, F. Mizuno, K. Tadanaga, T. Minami, M. Tatsumisago, Characterization of Li₂S–P₂S₅ glass-ceramics as a solid electrolyte for lithium secondary batteries, *Solid State Ionics* 175 (2004) 683–686, <https://doi.org/10.1016/j.ssi.2004.08.036>.
- [13] A. Sakuda, A. Hayashi, Y. Takigawa, K. Higashi, M. Tatsumisago, Evaluation of elastic modulus of Li₂S–P₂S₅ glassy solid electrolyte by ultrasonic sound velocity measurement and compression test, *J. Ceram. Soc. Japan* 121 (2013) 946–949, <https://doi.org/10.2109/jcersj2.121.946>.

- [14] Y. Li, D. Zhang, X. Xu, Z. Wang, Z. Liu, J. Shen, J. Liu, M. Zhu, Interface engineering for composite cathodes in sulfide-based all-solid-state lithium batteries, *J. Energy Chem.* 60 (2021) 32–60, <https://doi.org/10.1016/j.jechem.2020.12.017>.
- [15] A. Sakuda, T. Takeuchi, H. Kobayashi, Electrode morphology in all-solid-state lithium secondary batteries consisting of $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ and $\text{Li}_2\text{S-P}_2\text{S}_5$ solid electrolytes, *Solid State Ionics* 285 (2016) 112–117, <https://doi.org/10.1016/j.ssi.2015.09.010>.
- [16] G. Inoue, M. Kawase, Numerical and experimental evaluation of the relationship between porous electrode structure and effective conductivity of ions and electrons in lithium-ion batteries, *J. Power Sources* 342 (2017) 476–488, <https://doi.org/10.1016/j.jpowsour.2016.12.098>.
- [17] H. Nakamura, T. Kawaguchi, T. Masuyama, A. Sakuda, T. Saito, K. Kuratani, S. Ohsaki, S. Watano, Dry coating of active material particles with sulfide solid electrolytes for an all-solid-state lithium battery, *J. Power Sources* 448 (2020) 227579, <https://doi.org/10.1016/j.jpowsour.2019.227579>.
- [18] A. Sakuda, A. Hayashi, T. Ohtomo, S. Hama, M. Tatsumisago, All-solid-state lithium secondary batteries using LiCoO_2 particles with pulsed laser deposition coatings of $\text{Li}_2\text{S-P}_2\text{S}_5$ solid electrolytes, *J. Power Sources* 196 (2011) 6735–6741, <https://doi.org/10.1016/j.jpowsour.2010.10.103>.
- [19] K.H. Park, D.Y. Oh, Y.E. Choi, Y.J. Nam, L. Han, J.-Y. Kim, H. Xin, F. Lin, S.M. Oh, Y.S. Jung, Solution-processable glass $\text{LiI-Li}_4\text{SnS}_4$ superionic conductors for all-solid-state Li-ion batteries, *Adv. Mater.* 28 (2016) 1874–1883, <https://doi.org/10.1002/adma.201505008>.
- [20] B. Wang, Y. Zhao, M.N. Banis, Q. Sun, K.R. Adair, R. Li, T.-K. Sham, X. Sun, Atomic layer deposition of lithium niobium oxides as potential solid-state electrolytes for lithium-ion batteries, *ACS Appl. Mater. Interfaces* 10 (2018) 1654–1661, <https://doi.org/10.1021/acsami.7b13467>.
- [21] C. Wang, X. Li, Y. Zhao, M.N. Banis, J. Liang, X. Li, Y. Sun, K.R. Adair, Q. Sun, Y. Liu, F. Zhao, S. Deng, X. Lin, R. Li, Y. Hu, T.-K. Sham, H. Huang, L. Zhang, R. Yang, S. Lu, X. Sun, Manipulating interfacial nanostructure to achieve high-performance all-solid-state lithium-ion batteries, *Small Methods* 3 (2019) 1900261, <https://doi.org/10.1002/smt.201900261>.
- [22] H. Fathiannasab, A.G. Kashkooli, T. Li, L. Zhu, Z. Chen, Three-dimensional modeling of all-solid-state lithium-ion batteries using synchrotron transmission X-ray microscopy tomography, *J. Electrochem. Soc.* 167 (2020) 100558, <https://doi.org/10.1149/1945-7111/ab9380>.
- [23] H.-C. Yu, D. Taha, T. Thompson, N.J. Taylor, A. Drews, J. Sakamoto, K. Thornton, Deformation and stresses in solid-state composite battery cathodes, *J. Power Sources* 440 (2019) 227116, <https://doi.org/10.1016/j.jpowsour.2019.227116>.
- [24] A. Bielefeld, D.A. Weber, J. Janek, Microstructural modeling of composite cathodes for all-solid-state batteries, *J. Phys. Chem. C* 123 (2019) 1626–1634, <https://doi.org/10.1021/acs.jpcc.8b11043>.
- [25] A. Bielefeld, D.A. Weber, J. Janek, Modeling effective ionic conductivity and binder influence in composite cathodes for all-solid-state batteries, *ACS Appl. Mater. Interfaces* 12 (2020) 12821–12833, <https://doi.org/10.1021/acsami.9b22788>.
- [26] M. So, D. Naka, R. Goel, H. Yasui, Model development of a sponge carrier process using CFD-DEM with permeable particles, *J. Water Environ. Technol.* 10 (2012) 193–204, <https://doi.org/10.2965/jwet.2012.193>.
- [27] C. Sangrós Giménez, L. Helmers, C. Schilde, A. Diener, A. Kwade, Modeling the electrical conductive paths within all-solid-state battery electrodes, *Chem. Eng. Technol.* 43 (2020) 819–829, <https://doi.org/10.1002/ceat.201900501>.
- [28] T. Shi, Q. Tu, Y. Tian, Y. Xiao, L.J. Miara, O. Kononova, G. Ceder, High active material loading in all-solid-state battery electrode via particle size optimization, *Adv. Energy Mater.* 10 (2020) 1902881, <https://doi.org/10.1002/aenm.201902881>.
- [29] M. So, G. Inoue, R. Hirate, K. Nunoshita, S. Ishikawa, Y. Tsuge, Simulation of fabrication and degradation of all-solid-state batteries with ductile particles, *J. Electrochem. Soc.* 168 (2021), 030538, <https://doi.org/10.1149/1945-7111/abed23>.
- [30] M. So, G. Inoue, R. Hirate, K. Nunoshita, S. Ishikawa, Y. Tsuge, Effect of mold pressure on compaction and ion conductivity of all-solid-state batteries revealed by the discrete element method, *J. Power Sources* 508 (2021) 230344, <https://doi.org/10.1016/j.jpowsour.2021.230344>.
- [31] M. So, G. Inoue, K. Park, K. Nunoshita, S. Ishikawa, Y. Tsuge, Discrete Element Model for Deformation, Sintering, and Mold Compaction of Battery Electrodes, *MethodsX* (n.d.).
- [32] M. So, K. Park, T. Ohnishi, M. Ono, Y. Tsuge, G. Inoue, A discrete particle packing model for the formation of a catalyst layer in polymer electrolyte fuel cells, *Int. J. Hydrogen Energy* 44 (2019) 32170–32183, <https://doi.org/10.1016/j.ijhydene.2019.10.005>.
- [33] M. So, K. Park, Y. Tsuge, G. Inoue, A particle based ionomer attachment model for a fuel cell catalyst layer, *J. Electrochem. Soc.* 167 (2020), <https://doi.org/10.1149/1945-7111/ab68d4>, 013544.
- [34] M. So, M. Terashima, R. Goel, H. Yasui, Modelling the effect of biofilm morphology on detachment, *J. Water Environ. Technol.* 13 (2015) 49–62, <https://doi.org/10.2965/jwet.2015.49>.
- [35] L. Verlet, Computer “experiments” on classical fluids. I. Thermodynamical properties of Lennard-Jones molecules, *Phys. Rev.* 159 (1967) 98–103, <https://doi.org/10.1103/PhysRev.159.98>.
- [36] S.J. Cooper, A. Bertei, P.R. Shearing, J.A. Kilner, N.P. Brandon, TauFactor: an open-source application for calculating tortuosity factors from tomographic data, *Software* 5 (2016) 203–210, <https://doi.org/10.1016/j.softx.2016.09.002>.
- [37] E.J. Cheng, K. Hong, N.J. Taylor, H. Choe, J. Wolfenstine, J. Sakamoto, Mechanical and physical properties of $\text{LiNi}_{0.33}\text{Mn}_{0.33}\text{Co}_{0.33}\text{O}_2$ (NMC), *J. Eur. Ceram. Soc.* 37 (2017) 3213–3217, <https://doi.org/10.1016/j.jeurceramsoc.2017.03.048>.